Multimodal, Interactive Systems, UXD-M
Exercise, Summer term 2027

Exercise Sheet 01
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Toolchain, team and two working channels

What is this sheet about?

This sheet is the start of your semester project. You form the team that stays together until the end, fix a project idea, and get a first prototype running with two working channels. The modality vocabulary from the lecture – redundancy, complementarity, conflict, and the price tag on every channel – is what you use to justify which channel carries what.

Docking point: This exercise cashes in session T02 (*Channels with a price tag*): the modality vocabulary you sorted on the wall (redundancy, complementarity, conflict, and the price tag on every channel) now goes into a prototype you build with two working channels.

Organisation: Please work on this exercise sheet in teams of 3–4 students (at 15–20 students: four to six teams). The team you form today stays your project team for the whole semester; by the end of this block, **team and project idea are fixed**. Every team member needs a laptop if possible, at least two laptops per team. Webcam and microphone are required, so **check this before you sit down**.

AI role — tool (declared): You may generate, scaffold, and debug with any AI you like. Your upload must disclose what was made with which tool. The disclosure line is part of the submission, and a missing line counts as a missing submission.

Bonus points: This exercise track carries the objectives the written exam cannot test: building a working multimodal prototype and defending your design decisions in critique. In order for the bonus rule to apply (participation in all six blocks with an upload within 48 hours each earns 5 % compensation on the exam), your team must hand in one complete submission package (see the end of this sheet) within 48 hours after this exercise. Today's block is one of the six. A make-up block at the end of the semester is held in reserve; whether it takes place will be announced on Moodle. Throughout the track, **stick to the time budgets even when a task stays unfinished**: breadth before depth. The scope is set so that only very good teams finish. The sheet on Moodle is the binding basis; team size and submission scope can change from block to block.

First assignment (25 minutes): Form teams of 3–4 (plenum, then teams). Each team drafts one candidate project idea: an everyday task that plausibly benefits from two modalities. Your semester prototype will serve this task through at least two channels. Test the idea against the T02 question: where could the two channels contradict each other? **An idea without a possible conflict is too small.** If your idea stays vague, describe the everyday task with Golden Krishna's *CAT POET* questions first: **C**ontext, **A**ctors, **T**ask, **P**rocess you would like to simplify, **O**bjects in the environment, what needs to be **E**xplained, and which **T**riggers show that a person wants to act. The trigger line often names your second channel. And ask once: could a sensor (motion, pressure, proximity) remove one step of the process entirely, before any channel has to serve it? Write the idea down as one paragraph; you will revise it in the fourth assignment.

Second assignment (40 minutes): Toolchain smoke test, individually within the team. Every member, on their own machine, gets three tools running:

– *ProtoPie*: install the app, create an account, open a new project and make one trigger fire. Also open the voice-prototyping documentation (speech-to-text and text-to-speech triggers are built in; you will need them from exercise 3 on).

- *Figma Make*: Figma's prompt-to-prototype feature, browser only, Figma account required. Produce something — anything — from a one-sentence prompt.
- *MediaPipe Gesture Recognizer* from Google AI Edge runs as a web demo in the browser with your webcam, no install. The demo must recognize your open palm on webcam.

Team members help each other; the fast ones debug for the slow ones. **Nobody moves on to the third assignment until everyone has all three tools running.** Tool setup eats time in the least glamorous way possible, so it gets 40 minutes in the room.

Third assignment (55 minutes): Mini prototype with one interaction and two modalities (work in pairs within the team). Build the smallest thing that counts: one user intention (e.g. "confirm") served by two channels (e.g. a tap and a spoken "yes"), with visible feedback for both. Use whichever tool gets you there fastest; generated scaffolding is fine, **but you must be able to say what every part does**. You may throw this prototype away afterwards. Exercise block 4 asks you to wire two channels together again, at full size.

Fourth assignment (20 minutes): Cross-team demo and idea fixing (plenum). Each team shows its mini prototype in 90 seconds and reads out its project idea. The room asks exactly one question per team: *which of your two modalities is the one you would cut, and what does cutting it cost?* Afterwards, each team writes the final one-paragraph version of its idea (max 150 words).

Fifth assignment (10 minutes): Upload preparation. Assemble the submission package (see below) while everyone is still in the room. Teams that leave the packing for later usually end up doing it at hour 47 of the upload window.

Buffer and critique (30 minutes): Reserved for setup disasters (there will be some) and, if time remains, an open round: which tool surprised you, in which direction?

The submission (within 48 hours, one upload per team, via Moodle) consists of:

- `team.md` — team name, members, and the fixed project idea (one paragraph, max 150 words).
- `mini-prototype/` — the ProtoPie file, Figma Make link, or code of your mini prototype, plus a screen recording of both modalities working (max 60 seconds, .mp4, max 100 MB).
- `setup-proof.md` — one line per member confirming the three tools run, with a screenshot each (any format, max 5 MB per image).
- **AI disclosure line** (in `team.md`, mandatory): *"Made with AI: [tool] was used for [part]; checked and understood by [name]."* If no AI was used, write that; it is a disclosure either way.

Put everything into one upload per team and hand it in via Moodle within 48 hours after this exercise.